Enterprise System Architecture & Topology

High-concurrency open architecture, native Moodle core extension framework, and proposed cloud topology.

EDITORIAL SPECIFICATION

01



Student User Journey & Attainment Lifecycle

Nine-stage progressive lifecycle from public catalog discovery to automated digital certification.

STAGE 1: ONBOARDING & ADMISSION | STAGE 2: COMMERCE & ALLOCATION | STAGE 3: ACTIVE PEDAGOGY | STAGE 4: ATTAINMENT & COMPLETION

Autonomous End-to-End Learner Lifecycle — Zero manual administrative bottleneck between registration, enrolment, and faculty matching — EVENT-DRIVEN FLOW

01 CATALOG — Discover
Public course catalog & programs

- Guest / public view
- Syllabus & objectives
- Faculty credentials

TAG: COURSE DISCOVERY

02 ROLE CHECK — Register
Moodle core user creation

- Email verification flow
- Default role: Authenticated User
- GDPR / Terms consent

TAG: USER ONBOARDING

03 CURRICULUM — Select Course
Subject specialization & cohort

- Category navigation
- Cohort intake selection
- Fee schedule verification

TAG: COHORT SELECTION

04 FREE / PAID — Pay / Enrol
Gateway check or self-enrol

- Free: Instant enrol_self
- Paid: Razorpay / PhonePe
- Server verification check

TAG: ENROLMENT GATEWAY

05 COURSE ACCESS — Dashboard
Personalized learning cockpit

- Enrolled course view
- Syllabus & module navigation
- Triggers routing observer

TAG: LEARNER CONTEXT

06 ROUTING EVENT — Faculty Match
Automated Ola/Uber-Style Engine

- Specialization & subject match
- Real-time workload & capacity check
- Instant dual notification dispatch
- Fallback to pending retry queue

ACUITEY EXTENSION SERVICE

07 LTI 1.3 — Live Learning
Zoom LTI Pro virtual class

- OIDC secure seamless launch
- Live interactive lecture
- Recording archives in course
- Optional grade / attendance sync

TAG: SYNCHRONOUS CLASS

08 EVALUATION — Assess
Moodle native assessments

- Timed quizzes & question bank
- Assignment file submissions
- Faculty rubric grading
- Gradebook aggregation

TAG: ASSESSMENT SUITE

09 CERTIFICATE — Complete
Course completion milestone

- Activity criteria validation
- Automated digital certificate
- Verifiable credential hash
- Alumni cohort transition

TAG: FINAL ATTAINMENT

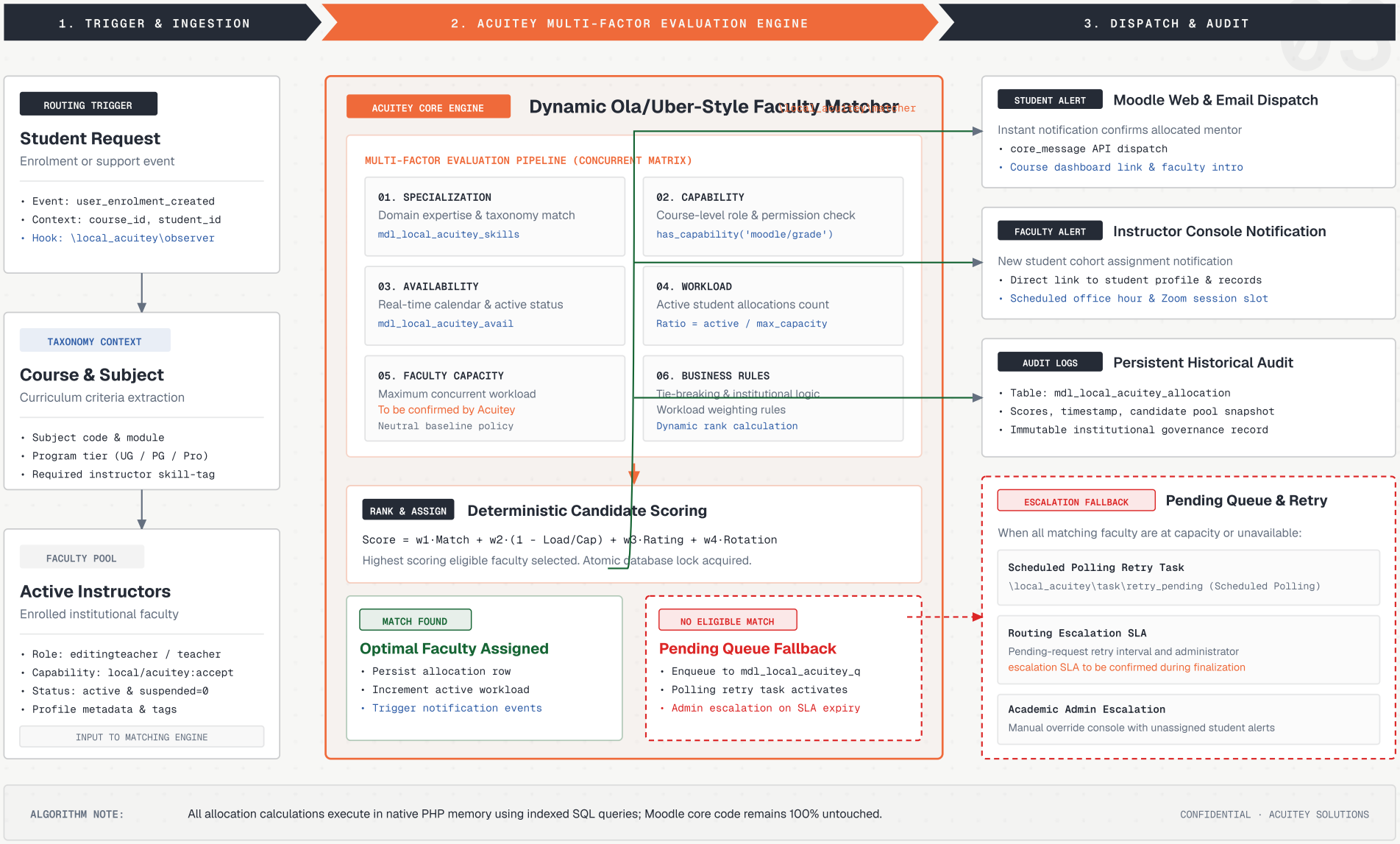
GOVERNANCE NOTE: Faculty allocation trigger to be confirmed: after payment, first activity launch, or explicit mentoring request.

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Automated Dynamic Faculty Routing Engine

Multi-factor concurrent evaluation pipeline: specialization, capability, availability, workload, and fallback.

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ALGORITHM NOTE: All allocation calculations execute in native PHP memory using indexed SQL queries; Moodle core code remains 100% untouched.

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Moodle Extension Architecture & Subsystem Boundary

Strict isolation of custom Acuitey logic within local_acuitey; Moodle 5.2 core code remains 100% untouched.

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04

ZONE A: MOODLE CORE SUBSYSTEMS (100% UNMODIFIED)

ZONE B: ACUITEY CUSTOM EXTENSION (local_acuitey)

UPSTREAM MOODLE

Core Platform Architecture

CORE UNTOUCHED

EVENTS API
\core\event*
Dispatches user, course, and enrolment triggers

CAPABILITIES
has_capability()
Granular contextual permission checking

USERS & AUTH
core_user / auth_*
Account lifecycles, sessions & user profiles

ENROLMENT
enrol_manual/self
Cohort allocations, expiry, course access

MESSAGING
core_message
Web notifications, email, and real-time alerts

TASK ENGINE
\core\task*
Adhoc queues & scheduled cron runners

EXTERNAL API
core_webservice
RESTful endpoints, mobile token service

LTI 1.3 ADVANTAGE
mod_lti
Zoom OIDC launch & NRPS roster provisioning

PAYMENT API
core_payment
Multi-gateway broker & payment accounts

Zero Core Modification Governance Principle

Moodle core code remains strictly pristine. Upstream point releases, security patches, and minor version updates apply cleanly via git with zero custom regression testing.

NO CODE PATCHES
/lib, /course, /user, /enrol files remain untouched

OFFICIAL API HOOKS
Uses supported events, tasks, and XMLDB abstractions

EVENT

AUTH

ALERT

CRON

ACUITEY EXTENSION

local_acuitey Custom Subsystem

Isolated Logic

FACULTY MATCHER
\local_acuitey\service\matcher
Multi-factor scoring algorithm and candidate selection

AVAILABILITY SERVICE
\local_acuitey\service\calendar
Real-time calendar checking & office hour availability status

WORKLOAD SERVICE
\local_acuitey\service\workload
Active student counter & capacity threshold evaluator

ROUTING ORCHESTRATOR
\local_acuitey\service\router
Coordinates pipeline, db writes, and dispatch triggers

METRICS & TELEMETRY
\local_acuitey\service\metrics
Match latency, queue size, and faculty distribution stats

ASSIGNMENT HISTORY
mdl_local_acuitey_alloc
Immutable historical ledger with score breakdowns

NOTIFICATION BRIDGE
\local_acuitey\service\notify
Formats & dispatches dual alerts via core_message

PENDING RETRY TASK
\local_acuitey\task\retry_pending
Scheduled polling task to re-evaluate queued requests

XMLDB CUSTOM SCHEMA (db/install.xml)

- mdl_local_acuitey_allocation
- mdl_local_acuitey_availability
- mdl_local_acuitey_queue
- mdl_local_acuitey_history

PLUGIN GOVERNANCE: Third-party plugins avoided to eliminate supply chain vulnerabilities; custom capabilities are fully self-contained in local_acuitey.

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Role & Security Model (Least Privilege Perimeter)

Spatial access control and capability matrix across five user personas and seven protected resource domains.

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05

ROLE PERSONAS & AUTHENTICATION → SPATIAL ACCESS ZONES & BOUNDARIES → PROTECTED RESOURCE DOMAINS

AUTHENTICATED PERSONAS

STUDENT — Learner Archetype

Enrolled Course context only

`moodle/course:view · mod/*:submit`

FACULTY — Teacher / Mentor

Allocated student cohorts only

`moodle/grade:view · mod_lti:launch`

ACADEMIC ADMIN

Curriculum & Faculty Matching

`local/acuitey:manage · moodle/course:update`

FINANCE ADMIN

Payment Audit & Settlement

`moodle/payment:manage · No grade access`

SUPER ADMIN

Platform Infrastructure & DevOps

`moodle/site:config · KVM / OS access`

PROTECTED RESOURCE DOMAINS (CONTEXT BOUNDARIES)

COURSES — Course Catalog & Modules

Syllabi, learning objects, video assets, forums, and LTI tools.

Access: Student (Read enrolled) · Faculty (Manage allocated)
Academic Admin (Full curriculum) · Finance (No access)

STUDENTS — Learner Profiles & PII

Identity, contact data, enrolment records, GDPR consents.

Access: Student (Self only) · Faculty (Allocated cohort only)
Cross-student record inspection strictly blocked by capability

ASSESSMENTS — Grades, Rubrics & Quizzes

Exam submissions, faculty feedback rubrics, and gradebook.

Access: Faculty (Grade assigned cohort) · Student (View own)
Finance Admin: ZERO VISIBILITY (Isolated by capability)

FINANCE — Payment Records & Payouts

core_payment logs, transaction IDs, gateway reconciliation.

Access: Finance Admin (Reconcile & audit) · Super Admin
Faculty / Students: ZERO ACCESS to institutional ledgers

ROUTING — Acuitey Routing & Queues

Dynamic allocation rules, pending retries, capacity limits.

Access: Academic Admin (Configure & override queue)
Faculty: View own queue · Student: Trigger via enrolment

REPORTING — Institutional Intelligence

Completion rates, faculty workloads, revenue summaries.

Access: Academic Admin (Pedagogy) · Finance (Revenue)
Super Admin (Consolidated telemetry across domains)

SYSTEM — System Infrastructure, Hostinger KVM & Security

Hostinger KVM environment, Nginx reverse proxy, PostgreSQL, Redis, Cloudflare Edge, and secret key storage.

Access: Super Admin / DevOps exclusively

Secrets never accessible via Moodle web interface

Moodle Contextual Access Control Equation

Every single database query and action executes strict context-level capability verification

MOODLE CONTEXT + CAPABILITY PERMISSION = AUTHORIZED SYSTEM ACTION

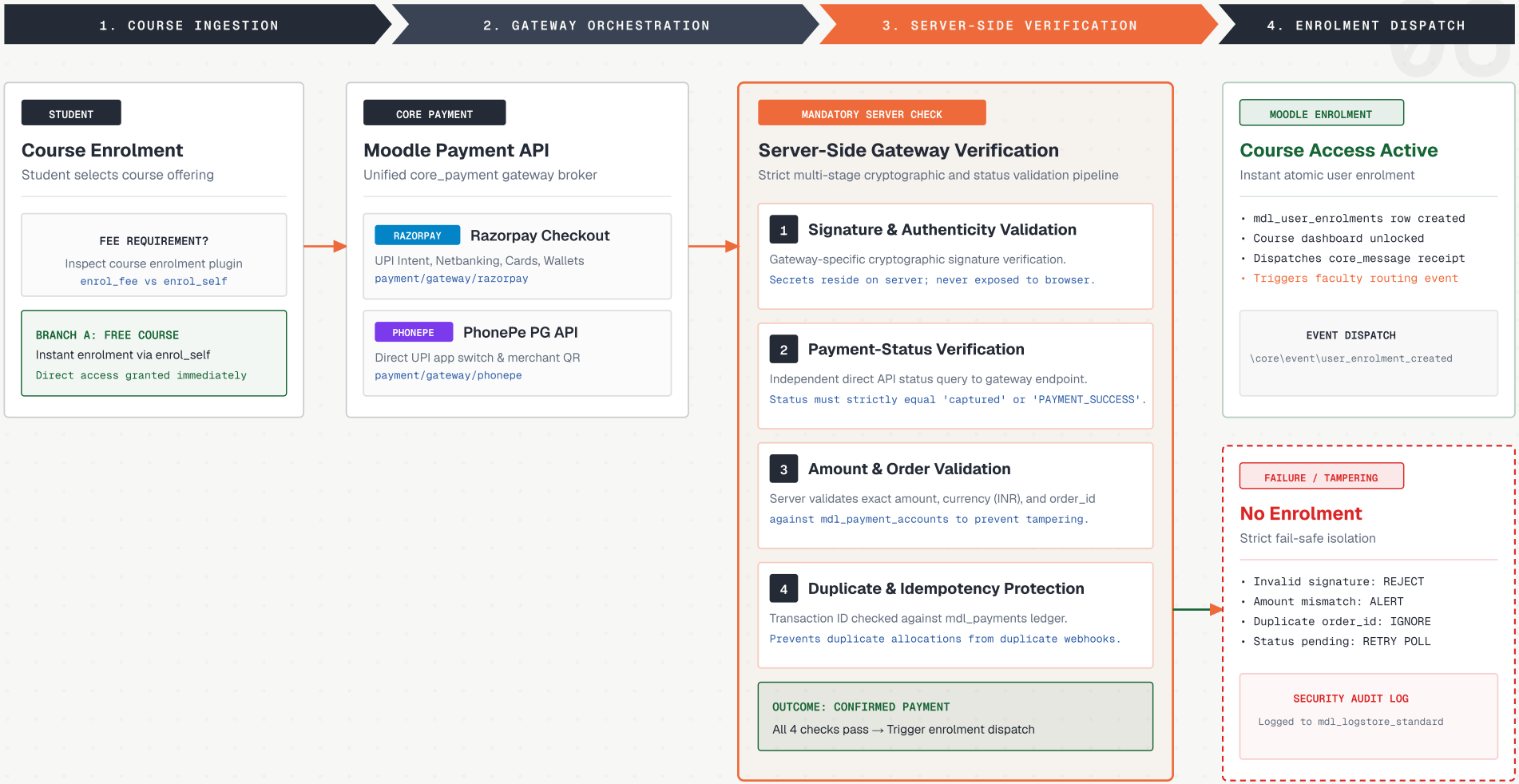
`require_capability()`

Dual Payment Gateway Architecture & Enrolment Flow

Razorpay and PhonePe integration via core_payment with mandatory server-side signature and status verification.

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06



COMMERCE GOVERNANCE: Confirm whether Razorpay and PhonePe will both be offered simultaneously and obtain required production merchant/API credentials.

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Zoom LTI 1.3 Advantage Integration Architecture

Cryptographic OIDC handshake, NRPS roster provisioning, and conditional attendance synchronization.

1. PARTICIPANT INITIATION | 2. LTI 1.3 ADVANTAGE OIDC PROTOCOL | 3. ZOOM MEETING ROOM

MOODLE PLATFORM BOUNDARY

STUDENT & FACULTY

Authenticated User

Active authenticated session

Browser context / Mobile app

COURSE ACTIVITY

Zoom LTI 1.3 Activity

Moodle native mod_lti module

- Click 'Join Virtual Class'
- Context: course_id, user_id

SECURITY ENGINE

LTI 1.3 Key Management

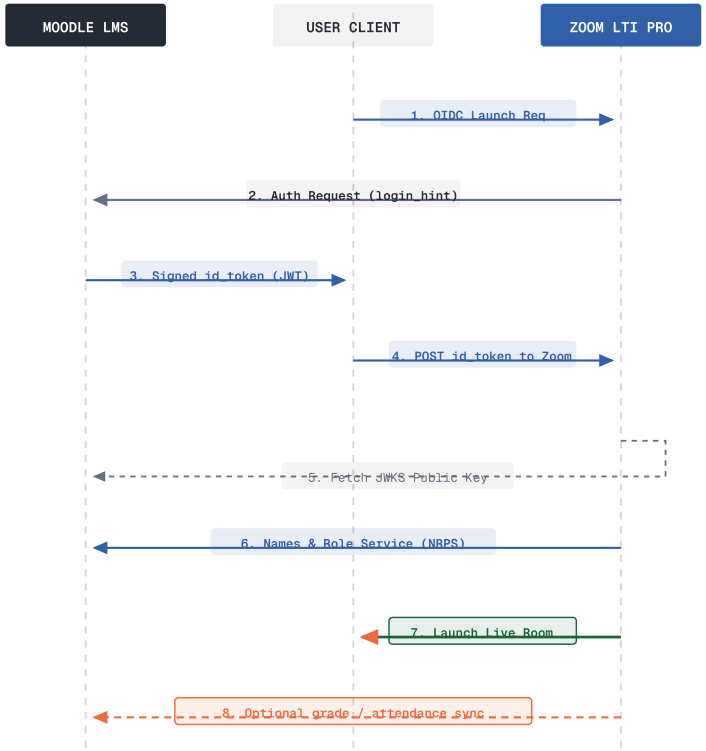
RSA-256 asymmetric keypair

- Private Key in mdl_lti_types
- Public JWKS endpoint exposed
- Signs JWT token payload
- Transmits nonce & state

STANDARDS COMPLIANCE

1EdTech LTI 1.3 Advantage Certified

CRYPTOGRAPHIC HANDSHAKE & LAUNCH SEQUENCE



ZOOM CLOUD INFRASTRUCTURE

LIVE CLASS

Zoom Meeting Session

Host role mapped to assigned faculty

- Audio / Video / Screen sharing

ROSTER ROLES

NRPS Role Mapping

Automatic role determination

- Faculty = Meeting Host / Moderator
- Student = Attendee / Participant

ATTENDANCE POLICY

Conditional Sync Policy

Grade and attendance synchronization may be enabled where support

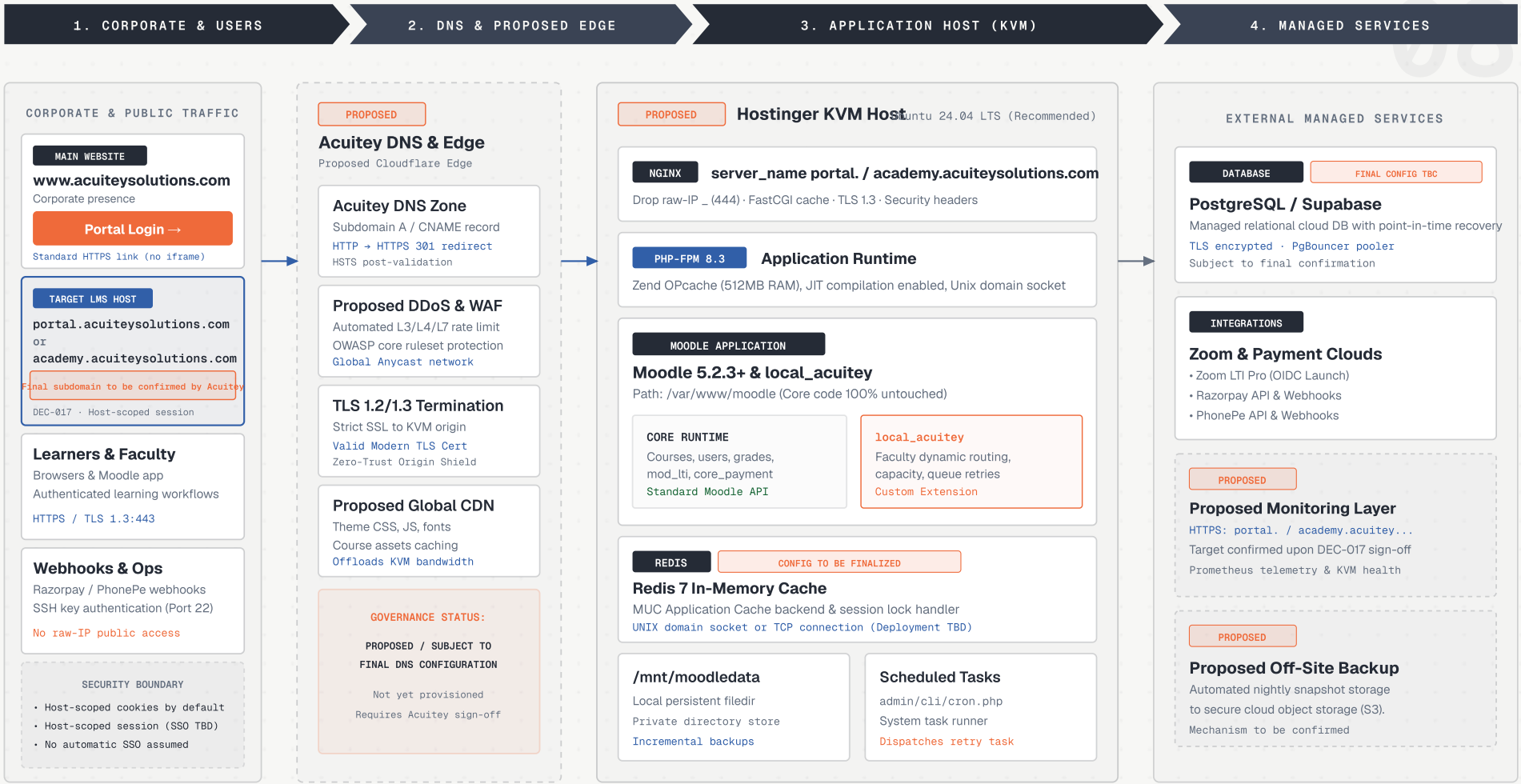
- Subject to Zoom plan type
- Pro vs Business account limits
- Webhook configuration required
- Not guaranteed prior to setup

ZOOM CONFIGURATION: Confirm Acuitey's Zoom account/licensing and the LTI Pro functions enabled for the account prior to production configuration.

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Proposed Production Deployment & Infrastructure Topology

Proposed Cloudflare Edge, Hostinger KVM, managed PostgreSQL, Redis cache, and telemetry layers.



INFRASTRUCTURE STATUS: Topology represents the recommended architecture; all cloud tiers are proposed pending final provisioning and merchant account sign-off.

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Implementation Roadmap & Architecture Confirmation

Eight-stage engineering delivery roadmap and formally confirmed institutional architecture declarations.

EDITORIAL SPECIFICATION

09

SECTION 1: IMPLEMENTATION ROADMAP (8 STAGES)

SECTION 2: ARCHITECTURE CONFIRMATION & DESIGN PRINCIPLES

CHRONOLOGICAL ENGINEERING ROADMAP

1	2	3	4	5	6	7	8
01 INFRA	02 CORE	03 EXTENSION	04 ROUTING	05 INTEGRATE	06 CONSOLE	07 AUDIT	08 LAUNCH
Cloud Setup	**Moodle 5.2**	**local_acuitey**	**Routing Engine**	**LTI & Pay**	**Dashboards**	**QA & Stress**	**Production**
Hostinger KVM provisioning, Ubuntu 24.04 OS baseline, PostgreSQL managed cloud, Redis deployment.	Moodle 5.2.3+ setup, PHP 8.3 OPcache, Nginx reverse proxy, TLS 1.3 hardening.	Plugin skeleton creation, XMLDB tables install, Event observers, Task subsystem bindings.	Matching algorithm, workload calculator, availability checker, pending queue retries.	Zoom LTI 1.3 Advantage, Razorpay gateway, PhonePe gateway, Server webhook security.	Academic routing admin, Finance audit screens, Prometheus telemetry, Observability dashboards.	Automated routing stress, Concurrency testing, Security vulnerability scan, Pilot cohort trial.	DNS cutover, Staff enablement, Go-live monitoring, SLA tracking.

ARCHITECTURE CONFIRMATION

Design Principles & Governance Declarations

MOODLE CORE INTEGRITY DECLARATION

> *"The Acuitey automated student-to-faculty workflow will be implemented using Moodle's supported extension architecture. Moodle core files will not be modified for this functionality. Moodle APIs, events, capabilities, tasks and messaging mechanisms will be used where appropriate, while Acuitey-specific routing logic will remain isolated within controlled custom extensions."*

THIRD-PARTY PLUGIN GOVERNANCE STATEMENT

> *"Third-party Moodle plugins will only be introduced where they provide a justified and maintainable capability that is not appropriately delivered through Moodle native functionality or the Acuitey custom extension."*

TARGET MOODLE RELEASE:	CUSTOM EXTENSION ID:	APPLICATION RUNTIME:	ARCHITECTURAL STATUS:
Moodle LMS 5.2.3+ (Build 20260916)	`local_acuitey (Isolated Subsystem)`	**PHP 8.3-FPM · OPcache · Hostinger KVM**	**Approved Initial System Design**

DOCUMENT CONFIRMATION: This specification establishes the agreed design principles for all Acuitey LMS engineering and integration activities.

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